A is an $m \times n$ real matrix and $b \in \mathbb{R}^m$

Recall setting: we are trying to solve the linear system $Ax = b$ but are worried it's inconsistent. So maybe try to instead find x such that Ax is as close to b as possible?

As usual think of A as representing a linear transformation from $\mathbb{R}^n$ to $\mathbb{R}^m$, what is point in range of A which is closest to b? That's another way to say it.

That is denoted $\hat{b}$, the projection of b on Col A. And then which $\hat{x} \in \mathbb{R}^n$ map to $\hat{b}$? Remember $\hat{b}$ is unique, but there might be many $\hat{x}$ which map to it.

We showed all $\hat{x}$ satisfying

$$A^T A\hat{x} = A^T b$$

are what is needed.

Turns out in most situations where we want to use this idea we need to find $\hat{x}$. Say want we to find a least squares curve, the question of whether there are multiple curves which accomplish that or not is relevant to us. So the question of whether or not $\hat{x}$ is unique can be quite important.

Boils down to: is $A^T A$ invertible or not?

If it is not, then $\hat{x}$ is not unique as $A^T Ax = A^T b$ is a consistent system where the coefficient matrix on the left, $A^T A$, is a square matrix which is not invertible, and that guarantees infinite solutions.

If $A^T A$ is invertible, then we can solve for $\hat{x}$ to get

$$\hat{x} = (A^T A)^{-1} A^T b$$

So then it is unique. Can we simplify this then perhaps? Maybe the first term, apply inverse of product idea?

Can't do that! Remember A is usually not square matrix so even though $A^T A$ is square, can't generally do that.

Hence we want to assess when $A^T A$ is invertible.

To check that, via Invertible Matrix Theorem, we want to assess whether

$$A^T A x = 0$$

has a nontrivial solution or not.

For this equation to hold we need

$$Ax \in \text{Nul}(A^T) = (\text{Col } A)^{\perp}$$

However $Ax \in \text{Col } A$ as well. So Ax has to be in the subspace $\text{Col } A$ as well as its orthogonal complement. As was shown in homework, this forces $Ax = 0$

So $A^T Ax = 0$ having a nontrivial solution is equivalent to whether $Ax = 0$ has a nontrivial solution.

And from what we learned earlier, the latter happens exactly when the columns of A are linearly dependent.

Thus we conclude:

$A^T A$ is invertible if and only if A has linearly independent columns.

Note again that don't say the last statement is equivalent to A being invertible - remember A is not necessarily square!

So $\hat{x}$ is unique exactly when A has linearly independent columns. Thus breaking A down in that setting makes sense. An algorithm we did recently which breaks down matrices with linearly independent columns is the QR Factorization. Turns out that can be useful here.

That is, remember that

$$\hat{x} = (A^T A)^{-1} A^T b$$

With this in mind, note via the QR Factorization of A:

$$A^T A = (QR)^T QR = R^T Q^T QR = R^T R$$

since Q has orthonormal columns in last step. Thus

$$\hat{x} = (R^T R)^{-1} (QR)^T b$$

$$= R^{-1}(R^T)^{-1}R^TQ^Tb$$
$$= R^{-1}Q^Tb$$

which might be an easier way to find $\hat{x}$. "Might" since it seems more convenient than the formula involving A, A^T, especially since R and Q should be easier matrices to work with than A usually. But you might have to compute Q, R from scratch from A so double edged.

But a good option to have.

Another way it can help is for again projection on Col A

$$\hat{b} = A\hat{x} = QR\hat{x} = QR(R^{-1}Q^Tb) = QQ^Tb$$

We mentioned earlier the formula UU^Tb for the projection, this is exactly that since Q has orthonormal columns, just a specific way of utilizing that formula.

6.5.15 For

$$A = \begin{bmatrix} 2 & 3 \\ 2 & 4 \\ 1 & 1 \end{bmatrix}, \quad b = \begin{bmatrix} 7 \\ 3 \\ 1 \end{bmatrix}$$

find $\hat{x}$, the point which maps to $\hat{b}$, the projection of b on Col A, using QR Factorization.

Columns of A being denoted as x_1, x_2 via Gram-Schmidt, $v_1 = x_1$ and then

$$v_2 = x_2 - \frac{x_2 \cdot v_1}{v_1 \cdot v_1} v_1 = \begin{bmatrix} 3 \\ 4 \\ 1 \end{bmatrix} - \frac{15}{9} \begin{bmatrix} 2 \\ 2 \\ 1 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} -1 \\ 2 \\ -2 \end{bmatrix}$$

So, normalizing, we get

$$Q = \frac{1}{3}\begin{bmatrix} 2 & -1 \\ 2 & 2 \\ 1 & -2 \end{bmatrix}$$

So then

$$R = Q^T A = \frac{1}{3}\begin{bmatrix} 2 & 2 & 1 \\ -1 & 2 & -2 \end{bmatrix}\begin{bmatrix} 2 & 3 \\ 2 & 4 \\ 1 & 1 \end{bmatrix}$$

$$= \frac{1}{3}\begin{bmatrix} 9 & 15 \\ 0 & 3 \end{bmatrix} = \begin{bmatrix} 3 & 5 \\ 0 & 1 \end{bmatrix}$$

From this,

$$R^{-1} = \frac{1}{3}\begin{bmatrix} -1 & 5 \\ 0 & 3 \end{bmatrix}$$

Thus finally,

$$x = R^{-1}Q^{T}b = \frac{1}{9}\begin{bmatrix} -1 & 5 \\ 0 & 3 \end{bmatrix}\begin{bmatrix} 2 & 2 & 1 \\ -1 & 2 & -2 \end{bmatrix}\begin{bmatrix} 7 \\ 3 \\ 1 \end{bmatrix}$$

---> $$x = \frac{1}{9}\begin{bmatrix} 7 & -8 & 11 \\ -3 & 6 & -6 \end{bmatrix}\begin{bmatrix} 7 \\ 3 \\ 1 \end{bmatrix}$$

---> $$= \frac{1}{9}\begin{bmatrix} 36 \\ -9 \end{bmatrix}$$

---> $$= \begin{bmatrix} 4 \\ -1 \end{bmatrix}$$

6.6 Design Matrix

We started last section by talking about least squares line, wondering how we compute it. But last section was really more about setting the stage for it. Now we will use those ideas to actually compute it as well as strategizing how to compute other "best fitting" curves.

Say in $\mathbb{R}^2$ we want least squares line for set of points:

$$(x_1, y_1), (x_2, y_2), \ldots, (x_n, y_n)$$

Suppose that line has equation $y = \beta_0 + \beta_1 x$ so our goal is naturally to "solve for" β_0, β_1 which make this line fit this set of points as best as possible.

In essence this is attempting to put all the mentioned points on the line in question.

As such the relevant equations become:

$$\mathcal{B}_0 + \mathcal{B}_1 x_1 = y_1$$

$$\mathcal{B}_0 + \mathcal{B}_1 x_2 = y_2$$

$$\vdots$$

$$\mathcal{B}_0 + \mathcal{B}_1 x_n = y_n$$

To help with this, take the following matrices:

$$X = \begin{bmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_n \end{bmatrix}, \quad \mathcal{B} = \begin{bmatrix} \mathcal{B}_0 \\ \mathcal{B}_1 \end{bmatrix}, \quad Y = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}$$

We call X the <u>design matrix</u> of this problem. Point being this system of equations is equivalent to the matrix equation:

$$X\mathcal{B} = Y$$

If this system was consistent, there would be a line through all these points, which would obviously be the least squares line.

But naturally that's not the interesting setting to us, and if $n > 2$ (certainly that would be typical), then it would be unlikely to happen.

So if system is inconsistent, find the point in range of X which is closest to Y. For that, would be the projection $\hat{Y}$ of Y on Col X. The value of $\mathcal{B}$, call it $\hat{\mathcal{B}}$, which creates that $\hat{Y}$ will correspond to the <u>least squares line</u> is the key.

Is this $\hat{\mathcal{B}}$ unique? Well presumably our set of points have distinct x-coordinates, guaranteeing X has linearly independent columns, ensuring the solution for $\hat{\mathcal{B}}$ is unique, i.e. least squares line is unique.
So how to find it? Same formula we discussed in the last section:

$$\hat{\mathcal{B}} = (X^T X)^{-1} X^T Y$$

2 x 2 so not that hard to find, even if set of points is unspecified. If you have seen a formula for least squares line, this is where it comes from.

Say same set of points we want best fitting quadratic (i.e. a parabola typically),

$$f(x) = \mathcal{B}_0 + \mathcal{B}_1 x + \mathcal{B}_2 x^2$$

How to find? Quite similar, use matrices

$$X = \begin{bmatrix} 1 & x_1 & x_1^2 \\ 1 & x_2 & x_2^2 \\ \vdots & \vdots & \vdots \\ 1 & x_n & x_n^2 \end{bmatrix}, \quad \mathcal{B} = \begin{bmatrix} \mathcal{B}_0 \\ \mathcal{B}_1 \\ \mathcal{B}_2 \end{bmatrix}$$

with same Y. Again relevant system is $X\beta = Y$

Can still usually use $\hat{\beta} = (X^T X)^{-1} X^T Y$, but that extra column for X will make these calculations much more extreme. Inverse becomes for 3 x 3, etc.

"Usually" because are we assured solution is unique still? Well now there are three columns in X instead of two so that's not nearly as clear. Indeed can find sets of points where it will no longer be unique

Ok most reasonable sets of points will still have it unique, but nevertheless, it's an additional concern that least squares line didn't really generally have.

Gave a question on worksheet of this type. In a sense the easiest version of such a question you could get, only four points (realistically the fewest for this as three or fewer would be on same parabola) with all points being clean.

And despite that, problem is still quite messy. That's issue with these problems - even if idea is similar when we go beyond line, it gets messy very quickly, like many things, not always practical to ask such questions to be done by hand.

But like many things important to understand how to do it even if it's not always practical to do by hand, especially if you want to teach a machine to do it.

6.7 Inner Product Spaces

In $\mathbb{R}^m$ or a subspace of it, if we wanted to express a vector as a linear combination of the basis elements for some basis, it was much easier if the basis was an orthogonal basis, have a nice formula to help with that.

Say we are working in vector space $\mathbb{P}_n$, the set of real polynomials of degree $\leq n$. If we have some basis for that (or some subspace of it), and we want to express a certain vector as a linear combination of those basis elements, might we have an easier time if that basis is an "orthogonal" basis?

But what does that even mean, the concept of orthogonal was only defined in $\mathbb{R}^m$ via dot product.

This is one motivation to expand that concept into what is called <u>inner product</u>.

Inner product of two vectors u and v, usually denoted as $\langle u, v \rangle$, is some real number. Ok in grand scheme of things, not always real number, but that's certainly the default in this class so assume that unless stated to be otherwise. Some properties inner product must satisfy:

(1) Commutative. I.e. for all u, v we have $\langle u, v \rangle = \langle v, u \rangle$
(2) $\langle u + v, w \rangle = \langle u, w \rangle + \langle v, w \rangle$
(3) For c any real scalar, $\langle cu, v \rangle = c\langle u, v \rangle$

So preserves addition and scalar multiplication. Also

(4) $\langle u, u \rangle \geq 0$, equalling 0 only when $u = 0$.

Same requirements that we had with dot product so in that sense not really anything new.

And logically it has to satisfy some properties, can't define it arbitrarily or then there won't be any structure endowed on the space so would not be meaningful.

First example is regular dot product on any $\mathbb{R}^m$

Next example, on say $\mathbb{R}^2$

$$<(x_1, y_1), (x_2, y_2)> = 2x_1 x_2 + 3y_1 y_2$$

Not necessarily a practically useful example but a good one to start with since it's a lot like regular dot product so not hard to see it satisfies the needed properties. That said, it is certainly different than regular dot product so dynamics do change, and it shows there are things beyond usual dot product.

On $\mathbb{P}_n$ take $n + 1$ distinct points $t_0, t_1, \ldots, t_n$

For any $p, q \in \mathbb{P}_n$ define

$$<p, q> = \sum_{i=0}^{n} p(t_i)q(t_i)$$

Easy to see this satisfies the early properties. But what about fourth one?

$$\langle p, p \rangle = \sum_{i=0}^{n} [p(t_i)]^2$$

This is ≥ 0 surely since sum of squares.

But why only = 0 when $p = 0$? Well that requires

$$\sum_{i=0}^{n} [p(t_i)]^2 = 0$$

which implies

$$p(t_0) = p(t_1) = \ldots = p(t_n) = 0$$

So p has at least $n + 1$ roots despite having degree at most n. Hence p being identically 0 here is indeed forced as needed.

This is why we needed $n + 1$ distinct points, with fewer you could have $< p, p > = 0$ without p being identically 0 which would defy this being an inner product.

So can define concept of orthogonal via this in $\mathbb{P}_n$ potentially. Take say the usual basis in $\mathbb{P}_n$

$$\{1, x, x^2, \ldots, x^n\}$$

Is this an orthogonal basis in this inner product?

After all the "standard basis" in $\mathbb{R}^m$ is an orthogonal basis (in fact orthonormal basis) under the usual dot product, that's part of what makes it so special. So it feels like this ought to be as well.

But it may not be here, it likely will depend on which $n + 1$ points you pick for the evaluation.

Bigger point is: if it is not, can we use Gram-Schmidt to make it so?

Indeed we can! That's key, those earlier ideas can be expanded into other vector spaces (well inner product spaces once such is defined), just by using inner product in place of dot product.

One big thing is to get accustomed to doing things like that in settings which are less familiar to us than $\mathbb{R}^m$.